

Computer Network Lab: 50 MCQs with Options and Answers and few short questions

1. Which device operates only at the physical layer of OSI model?

- a) Switch
- b) Router
- c) Hub
- d) Bridge

Answer: c) Hub

2. Which command is used to test network connectivity?

- a) tracert
- b) ping
- c) ipconfig
- d) nslookup

Answer: b) ping

3. Which protocol is used for echo request and reply?

- a) TCP
- b) UDP
- c) ICMP
- d) HTTP

Answer: c) ICMP

4. What does `tracert` command mainly show?

- a) MAC address
- b) IP conflict
- c) Path to destination
- d) Number of VLANs

Answer: c) Path to destination

5. What is the IP address range for Class C networks?

- a) 0-127
- b) 128-191
- c) 192-223
- d) 224-239

Answer: c) 192-223

6. What is the main disadvantage of a hub?

- a) Intelligent data filtering
- b) Data flooding
- c) Secure communication
- d) Speed boost

Answer: b) Data flooding

7. Which device uses MAC addresses to forward frames?

- a) Router
- b) Switch
- c) Gateway
- d) Hub

Answer: b) Switch

8. Which device connects different networks with different protocols?

- a) Switch
- b) Hub
- c) Bridge
- d) Gateway

Answer: d) Gateway

9. At which layer does a router primarily operate?

- a) Layer 1
- b) Layer 2
- c) Layer 3
- d) Layer 7

Answer: c) Layer 3

10. Which device amplifies signals and acts like a repeater?

- a) Passive hub
- b) Bridge
- c) Active hub
- d) Switch

Answer: c) Active hub

11. Which command assigns ports to a VLAN?

- a) switchport mode trunk
- b) switchport access vlan
- c) vlan create
- d) interface trunk enable

Answer: b) switchport access vlan

12. What does a trunk port do?

- a) Carries traffic from only one VLAN
- b) Carries traffic from multiple VLANs
- c) Blocks VLAN traffic
- d) Acts as firewall

Answer: b) Carries traffic from multiple VLANs

13. What is VLAN 1 usually called?

- a) Default VLAN
- b) Management VLAN
- c) Trunk VLAN
- d) Access VLAN

Answer: a) Default VLAN

14. Which command shows VLAN configurations on a switch?

- a) show vlan
- b) show vlan brief
- c) show trunk
- d) show port

Answer: b) show vlan brief

15. Why is VLAN used?

- a) Increase speed
- b) Separate broadcast domains
- c) Disable MAC addresses
- d) Amplify signals

Answer: b) Separate broadcast domains

16. Which command brings an interface up?

- a) shutdown
- b) no shutdown
- c) enable port
- d) start interface

Answer: b) no shutdown

17. Which command shows the routing table in a router?

- a) show vlan
- b) show interface
- c) show ip route
- d) show routes

Answer: c) show ip route

18. What is the first step in router configuration?

- a) enable
- b) show ip route
- c) ping
- d) copy running-config startup-config

Answer: a) enable

19. Which command saves the current configuration in a router?

- a) save config
- b) write

- c) copy running-config startup-config
- d) save running

Answer: c) copy running-config startup-config

20. Which of the following is a static route command?

- a) ip route
- b) router ospf
- c) switchport mode
- d) trunk route

Answer: a) ip route

21. What is the mathematical model used for inter-arrival times?

- a) Uniform distribution
- b) Normal distribution
- c) Negative exponential distribution
- d) Poisson distribution

Answer: c) Negative exponential distribution

22. What Excel formula is used to count arrivals up to a certain time?

- a) SUMIF
- b) COUNTIF
- c) AVERAGEIF
- d) VLOOKUP

Answer: b) COUNTIF

23. Traffic intensity A is given by:

- a) λ / T
- b) $\lambda \times T$
- c) $\lambda + T$
- d) $\lambda - T$

Answer: b) $\lambda \times T$

24. Which chart shows the probability distribution in traffic simulation?

- a) Line chart
- b) Bar graph
- c) Pie chart
- d) Scatter plot

Answer: b) Bar graph

25. Which traffic model is based on random arrival?

- a) TCP traffic
- b) Poisson process
- c) UDP traffic
- d) HTTP traffic

Answer: b) Poisson process

26. What is the default port number for HTTP?

- a) 20
- b) 25
- c) 80
- d) 443

Answer: c) 80

27. Which command creates a TCP socket in Python?

- a) `socket.socket(AF_INET, SOCK_STREAM)`
- b) `socket.create(AF_INET, SOCK_STREAM)`
- c) `socket.new(AF_INET, SOCK_DGRAM)`
- d) `socket(AF_INET, SOCK_STREAM)`

Answer: a) `socket.socket(AF_INET, SOCK_STREAM)`

28. What function listens for incoming connections?

- a) `connect()`
- b) `accept()`
- c) `listen()`
- d) `bind()`

Answer: c) `listen()`

29. Which function is used by client to connect to server?

- a) accept()
- b) send()
- c) connect()
- d) bind()

Answer: c) connect()

30. Which socket type is connection-oriented?

- a) Stream socket (TCP)
- b) Datagram socket (UDP)
- c) Raw socket
- d) IP socket

Answer: a) Stream socket (TCP)

31. What port number is used for POP3?

- a) 80
- b) 25
- c) 110
- d) 443

Answer: c) 110

32. What information does a DHCP server provide?

- a) Username and password
- b) IP address and gateway
- c) Switchport address
- d) VLAN name

Answer: b) IP address and gateway

33. Which protocol is used to dynamically allocate IP addresses?

- a) DNS
- b) HTTP
- c) DHCP
- d) SMTP

Answer: c) DHCP

34. Which Wi-Fi security protocol is stronger?

- a) WEP
- b) WPA2
- c) WPA
- d) Open

Answer: b) WPA2

35. What does SSID stand for?

- a) Secure System ID
- b) Secure Set Internet Domain
- c) Service Set Identifier
- d) Server Set Identification

Answer: c) Service Set Identifier

36. What type of routing protocol is OSPF?

- a) Distance-vector
- b) Link-state
- c) Path-vector
- d) Source-routing

Answer: b) Link-state

37. What does OSPF use to elect a designated router?

- a) IP address
- b) Hello protocol
- c) RIP protocol
- d) LSA

Answer: b) Hello protocol

38. Which mask is used with OSPF network statements?

- a) Subnet mask
- b) Wildcard mask

- c) Network mask
- d) Host mask

Answer: b) Wildcard mask

39. Which router command starts OSPF process 1?

- a) router ospf 1
- b) router ospf process1
- c) ospf enable 1
- d) router process 1

Answer: a) router ospf 1

40. Which area is usually called OSPF backbone?

- a) Area 5
- b) Area 0
- c) Area 1
- d) Area 10

Answer: b) Area 0

41. What is the IPv4 loopback address?

- a) 0.0.0.0
- b) 192.168.0.1
- c) 127.0.0.1
- d) 10.0.0.1

Answer: c) 127.0.0.1

42. Which layer is responsible for packet delivery?

- a) Network
- b) Data Link
- c) Transport
- d) Application

Answer: a) Network

43. Which layer is responsible for end-to-end communication?

- a) Application
- b) Transport
- c) Network
- d) Data Link

Answer: b) Transport

44. Which address is 48 bits long?

- a) IP address
- b) MAC address
- c) Port address
- d) Host address

Answer: b) MAC address

45. What is the name of router-on-a-stick configuration?

- a) VLAN Trunking
- b) Sub-interface Routing
- c) Static Routing
- d) Inter-Switch Link

Answer: b) Sub-interface Routing

46. Which device is called a multiport repeater?

- a) Switch
- b) Hub
- c) Bridge
- d) Gateway

Answer: b) Hub

47. What is the purpose of the Accept() function in server socket programming?

- a) Bind the server
- b) Listen to incoming connection
- c) Accept an incoming connection
- d) Connect to another server

Answer: c) Accept an incoming connection

48. How many VLANs are configured in your lab?

- a) 2
- b) 3
- c) 4
- d) 5

Answer: c) 4

49. Which port number is used for SMTP?

- a) 25
- b) 110
- c) 143
- d) 53

Answer: a) 25

50. Which type of IP address is dynamically allocated?

- a) Static IP
- b) Manual IP
- c) DHCP IP
- d) Reserved IP

Answer: c) DHCP IP

Short Questions with Answers:

Routing, Hub, Switch, Router, DHCP

What is the purpose of using a cross cable in a network setup?

→ To connect two similar devices (like PC to PC or Switch to Switch).

Why is a static IP address used instead of DHCP in some networks?

→ Because there is no DHCP server available.

Define ICMP.

→ Internet Control Message Protocol, used for network diagnostics.

Write an example of a Class C IP address.

→ 192.168.25.37

What command is used to trace the route to a specific IP address?

→ tracer IP_address

What does the ping command test?

→ Connectivity between two network devices.

Mention one disadvantage of using a hub.

→ Causes network flooding due to broadcasting.

What is an active hub?

→ A hub that amplifies the incoming signal (needs power).

At which OSI layer does a bridge operate?

→ Data Link Layer (Layer 2).

How does a layer-2 switch forward frames?

→ Based on the destination MAC address.

Which switch can route packets based on IP addresses — Layer-2 or Layer-3?

→ Layer-3 switch.

What is the main function of a router?

→ To connect different networks.

What is a gateway in networking?

→ A device that translates communication between different protocols.

State one difference between a router and a layer-3 switch.

→ Router uses software-based routing; Layer-3 switch uses hardware-based routing.

What does the command tracer /? show?

→ Help options for tracer command.

Which command is used to enter privileged EXEC mode in a router?

→ enable (short form: en).

Write the command to configure IP address 192.168.1.2 with subnet mask 255.255.255.0 on interface fa0/0.

→ ip address 192.168.1.2 255.255.255.0

What does no shutdown command do?

→ It activates the interface.

How do you view routing information in a router?

→ Using show ip route.

What is the purpose of the ip route command?

→ To configure static routes.

What is a VLAN?

→ A logically separated network within a switch.

What command is used to create VLAN 10 with name 'A'?

→ vlan 10 then name A

Which command assigns ports fa0/1 to fa0/6 to VLAN 10?

→ interface range fa0/1-6 then switchport access vlan 10

What does switchport mode trunk do?

→ Enables the interface to carry multiple VLANs.

In a VLAN setup, why do we use sub-interfaces on a router?

→ To route traffic between VLANs (Router-on-a-Stick).

Give an example of an encapsulation command for VLAN 10.

→ encapsulation dot1q 10

Write the IP address and gateway for VLAN 20 based on the lab.

→ IP: 192.168.20.2, Gateway: 192.168.20.1

What is the default VLAN ID?

→ VLAN 1.

Which VLAN connects PC with IP address 192.168.30.2?

→ VLAN 30.

What is a socket?

→ An endpoint for sending or receiving data across a network.

Name two types of sockets.

→ Stream Socket (TCP) and Datagram Socket (UDP).

What does .AF_INET represent in socket programming?

→ IPv4 address family.

What is the role of the bind() function in socket programming?

→ It binds a socket to an IP address and port.

What Python function is used to establish a connection to a server?

→ connect().

Write the default port number for HTTP communication.

→ Port 80.

What is the purpose of accept() function on the server side?

→ To accept an incoming connection request.

In socket communication, which function sends data?

→ send() or sendall().

Which function is used to receive data from the socket?

→ recv().

Mention one difference between TCP and UDP socket.

→ TCP is connection-oriented; UDP is connectionless.

What port does HTTP use by default?

→ Port 80.

What information should be stored on the DNS server based on the lab?

→ DNS name and corresponding IP address.

Mention two HTML tags used in the webpage loaded on the DNS server.

→ <title>, <h1>.

Write the URL used to browse the JU server webpage.

→ <http://www.ju.edu/ju.html>

What is the function of the clock rate command in a serial connection?

→ Sets the data transmission rate (used by DCE).

What is SSID in wireless LAN?

→ The network name (Service Set Identifier).